STOCKSENSE

PROJECT RISK REGISTER

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Phase: Month 1 - Week 3 Deliverable

> **Risk Management Strategy:** This Risk Register identifies, assesses, and outlines mitigation protocols for technical, supply chain, environmental, and project schedule risks across the 6-month lifecycle of StockSense.

1. TECHNICAL & ENGINEERING RISKS

RISK ID	RISK EVENT & DESCRIPTION	IMPACT	LIKELIHOOD	MITIGATION & CONTINGENCY PLAN
TR-01	**Signal Drift & Vibration Noise:** Mechanical vibration or thermal shifts cause ADC count instability.	HIGH	MEDIUM	Implement median + moving average digital filtering in firmware; integrate dynamic software temperature compensation.
TR-02	**Network Disconnection Data Loss:** Prolonged Wi-Fi failure leads to dropped telemetry packets.	HIGH	LOW	Utilize onboard SPI Flash memory ring buffer for offline queueing; auto-flush via MQTT on connection recovery.
TR-03	**Power Path Switchover Reset:** Mains outage causes MCU brownout before battery failover completes.	HIGH	MEDIUM	Design PCB power path using ideal diode circuit (TP4056 + MOSFET) and decoupling caps for zero-reset transition.
TR-04	**Load Cell Creep & Mechanical Hysteresis:** Static bin loads degrade weight accuracy over time.	MEDIUM	MEDIUM	Select rated high-grade strain gauges; implement periodic automatic zero-point re-calibration logic during empty bin states.

2. SUPPLY CHAIN & PROCUREMENT RISKS

RISK ID	RISK EVENT & DESCRIPTION	IMPACT	LIKELIHOOD	MITIGATION & CONTINGENCY PLAN
SR-01	**Component Import Lead Times:** Primary ICs (ESP32-WROOM, HX711, PN532) delayed in transit.	HIGH	HIGH	Finalize BOM early in Month 1; pre-order hardware parts; select pin-compatible back-up alternatives (ADS1232, MFRC522).
SR-02	**BOM Budget Overrun:** Prototype manufacturing costs exceed target bounds (> LKR 12,000).	HIGH	MEDIUM	Audit component suppliers in Month 2; leverage bulk local purchasing and optimize passives on custom PCB layer.
SR-03	**Enclosure 3D Printing Bottlenecks:** Local 3D printing equipment delays physical bin platform.	MEDIUM	MEDIUM	Finalize CAD designs in FreeCAD early (Month 2, Week 6); run test fits on basic PLA before final PETG prints.

3. RISK ASSESSMENT MATRIX OVERVIEW

10	6	3	100%
TOTAL TRACKED RISKS	HIGH SEVERITY	MEDIUM SEVERITY	MITIGATION COVERED

4. ENVIRONMENTAL & OPERATING RISKS

RISK ID	RISK EVENT & DESCRIPTION	IMPACT	LIKELIHOOD	MITIGATION & CONTINGENCY PLAN
ER-01	**High Ambient Humidity / Moisture:** Condensation causes circuit shorts or corrosion in tropical sites.	HIGH	HIGH	Apply conformal coating to assembled PCB; design IP54-rated splash-resistant 3D enclosure with silicone seals.
ER-02	**Thermal Overheating:** High shop-floor heat (> 45°C) degrades battery performance or MCU stability.	MEDIUM	MEDIUM	Integrate thermal monitoring via I²C sensor; add thermal relief pads and passive ventilation louvers in enclosure.
ER-03	**Physical Impact & Overload:** Operators drop heavy items exceeding load cell capacity rating.	HIGH	MEDIUM	Incorporate mechanical mechanical overload stops in 3D frame structure to physically prevent load cell bending past limit.

5. SCHEDULE & EXECUTION RISKS

RISK ID	RISK EVENT & DESCRIPTION	IMPACT	LIKELIHOOD	MITIGATION & CONTINGENCY PLAN
SCH-01	**PCB Revision Iteration Delay:** Mistakes in KiCad PCB v1 force a secondary fabrication cycle.	HIGH	MEDIUM	Conduct comprehensive DRC/DFM checks in KiCad; perform peer schematic review before ordering Gerber files.
SCH-02	**Integration Bottleneck:** Delays connecting MQTT edge nodes to backend cloud API and React UI.	MEDIUM	MEDIUM	Freeze MQTT topic tree and JSON payload structures in Week 4; use simulated Wokwi edge nodes for early frontend testing.

Review Protocol: This Risk Register will be re-evaluated bi-weekly during Friday progress meetings and formally updated at each Gate Review (Gate 1: End of Month 2, Gate 2: End of Month 4, Gate 3: Mid Month 6).